EXP NO:4
7/8/26

Packet Sniffing using raw sockets in Python.

AIM:

To implement a packet sniffer using raw sockets in python that captures live network traffic and parses ethernet, IP, and transport-layer (TCP/UDP, ICMP) headers.

STEPS:

STEP1 : Set up the lab environment
STEP2 : Understand the packet structure you Phrase
STEP3 : Create the raw socket
STEP4 : Parse the ethernet Header
STEP5 : Parse the IP header
STEP6 : Parse TCP/UDP/ICMP Headers
STEP7 : Write the main sniffing loop
STEP8 : Run the sniffer
STEP9 : Extend the lab

RESULT:

Thus, a packet sniffer using raw sockets that captures live network traffic and parses ethernet, IP and transport-layer header implement and achieved successfully.

ne.py

```
import socket
import struct
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet Sniffer Started...\n")
try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)
        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])
        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination :", destination)
        print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Packet Sniffer Started...

----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.168.56.1
Destination : 224.0.0.251
Packet Length: 56
----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.16[illegible]6.1
Destination : 224.[illegible]51
```